



IT SIMPLERA INSTITUTE INTERNSHIP PROGRAM

WEEK 03 REPORT

Malware Detection and Incident Response using pfSense, Squid Proxy, ClamAV, Syslog-NG, and Wazuh SIEM

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1. Introduction

Malware is one of the most common cyber threats affecting modern organizations. Detecting malicious files and monitoring suspicious network activity are essential responsibilities of a Security Operations Center (SOC). Security tools such as firewalls, proxy servers, antivirus solutions, and SIEM platforms work together to prevent, detect, and respond to security incidents.

In this project, pfSense Firewall was used to secure the network, Squid Proxy was configured to inspect web traffic, ClamAV was integrated to scan files for malware, and Wazuh SIEM was configured to monitor security events. The EICAR test file was used to safely simulate malware detection, while Wazuh generated security alerts that were analyzed to understand the complete incident response workflow.

2. Objectives

The objectives of this project were to:

- Deploy and configure the pfSense Firewall.
 - Install and configure Squid Proxy.
 - Integrate ClamAV with Squid Proxy.
 - Configure Syslog-NG to forward logs to Wazuh.
 - Integrate pfSense logs with the Wazuh SIEM.
 - Create custom Wazuh decoders and detection rules.
 - Test malware detection using the EICAR test file.
 - Monitor and analyze security alerts.
 - Develop an Incident Response Plan.
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3. Software Requirements

The following software and tools were used during the implementation:

- VMware Workstation
- pfSense Firewall
- Squid Proxy
- ClamAV Antivirus
- Syslog-NG

- Wazuh SIEM
- Kali Linux
- Windows 10
- Web Browser

4. Deploying pfSense Firewall

The pfSense Firewall virtual appliance was successfully deployed inside VMware Workstation. Network interfaces were configured, and the firewall was initialized using the setup wizard. WAN and LAN interfaces were verified before proceeding with additional security configurations.

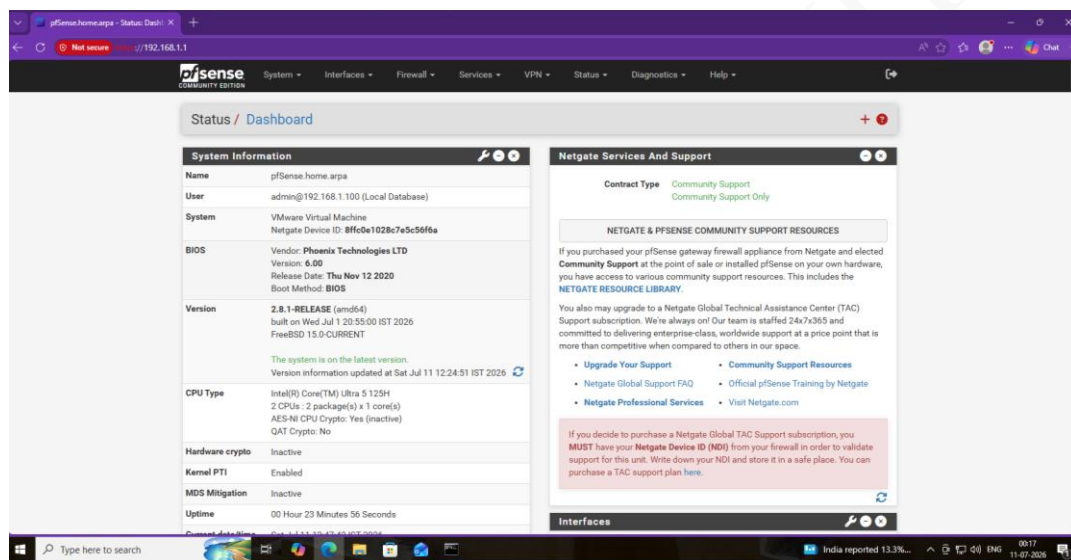


Figure 1: pfSense Firewall successfully deployed and configured.

5. Installing and Configuring Squid Proxy

The Squid Proxy package was installed through the pfSense Package Manager. The proxy service was enabled and configured to inspect web traffic passing through the firewall. The proxy server acts as an intermediary between client systems and external web servers, improving traffic control and security monitoring.

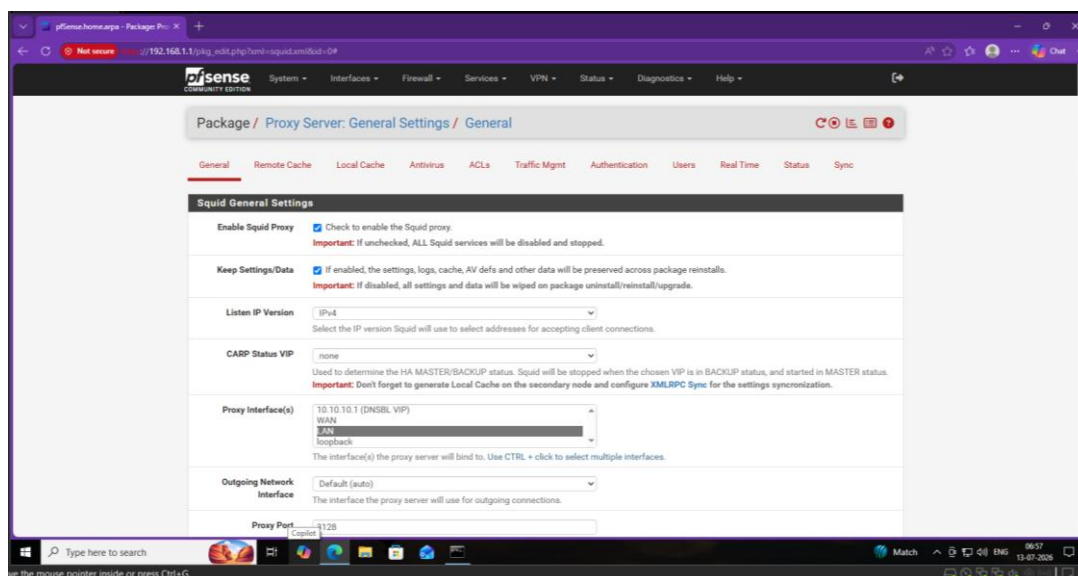


Figure 2: Squid Proxy successfully installed and configured.

6. Installing and Configuring ClamAV

ClamAV was integrated with Squid Proxy to perform antivirus scanning on downloaded files and web content. The antivirus engine was updated with the latest malware signatures, enabling the detection of malicious files before they reached client systems.

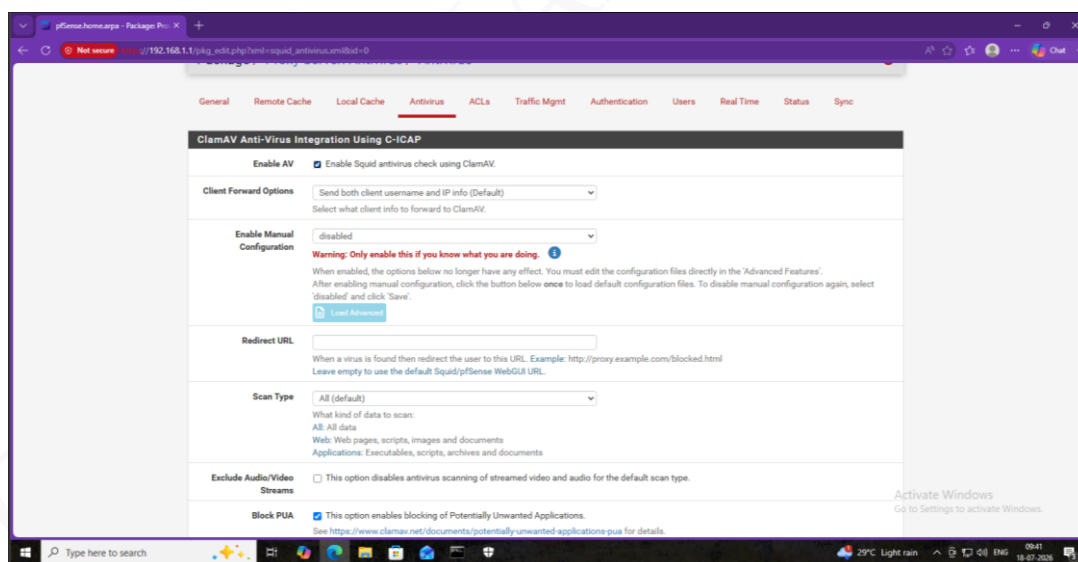


Figure 3: ClamAV integrated with Squid Proxy for malware scanning.

7. Configuring Syslog-NG

Syslog-NG was configured to forward firewall and proxy logs from pfSense to the Wazuh SIEM server. This enabled centralized log collection and provided real-time visibility into firewall activities and security events.

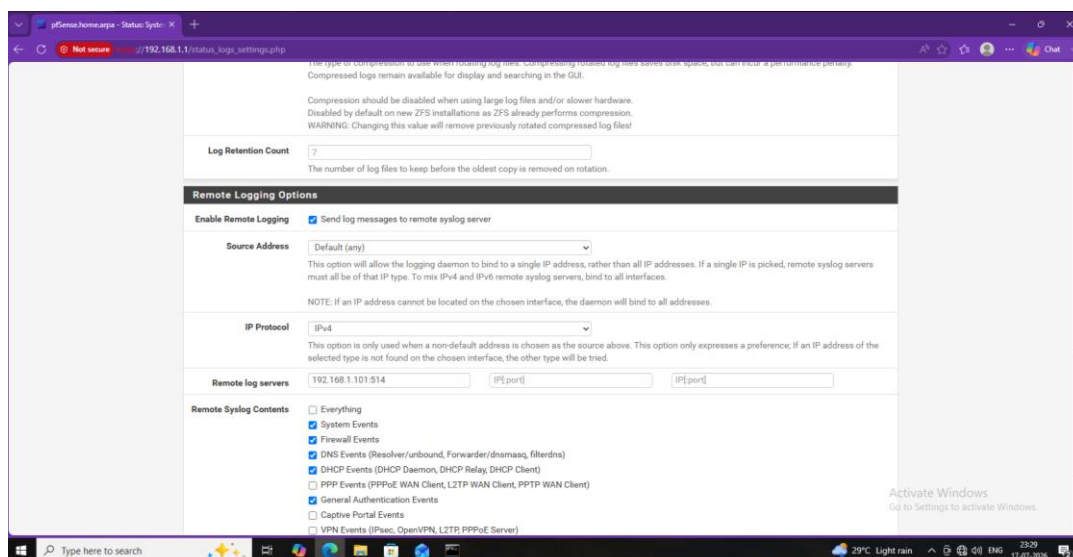


Figure 4: Syslog-NG configured to forward logs to Wazuh SIEM.

8. Wazuh SIEM Integration

The Wazuh SIEM platform was configured to receive and process logs generated by pfSense. Log forwarding was verified successfully, allowing firewall events and proxy activities to be monitored from a centralized dashboard.

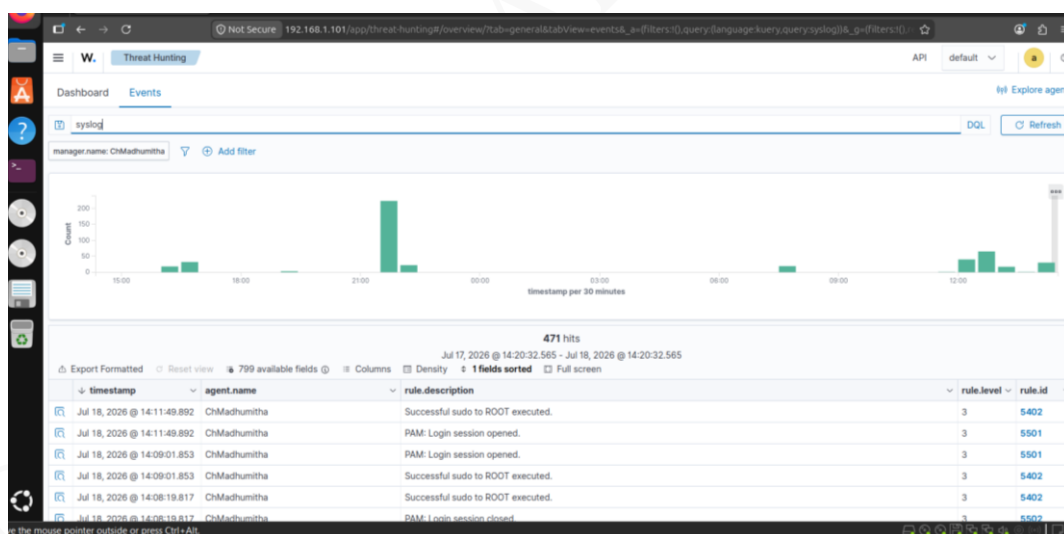


Figure 5: Wazuh Dashboard Monitoring Security Events

9. Creating Custom Wazuh Decoders and Rules

Custom decoders and detection rules were created to improve log parsing and identify malware-related events. These custom configurations enabled Wazuh to recognize specific log patterns and generate meaningful security alerts for suspicious activities.

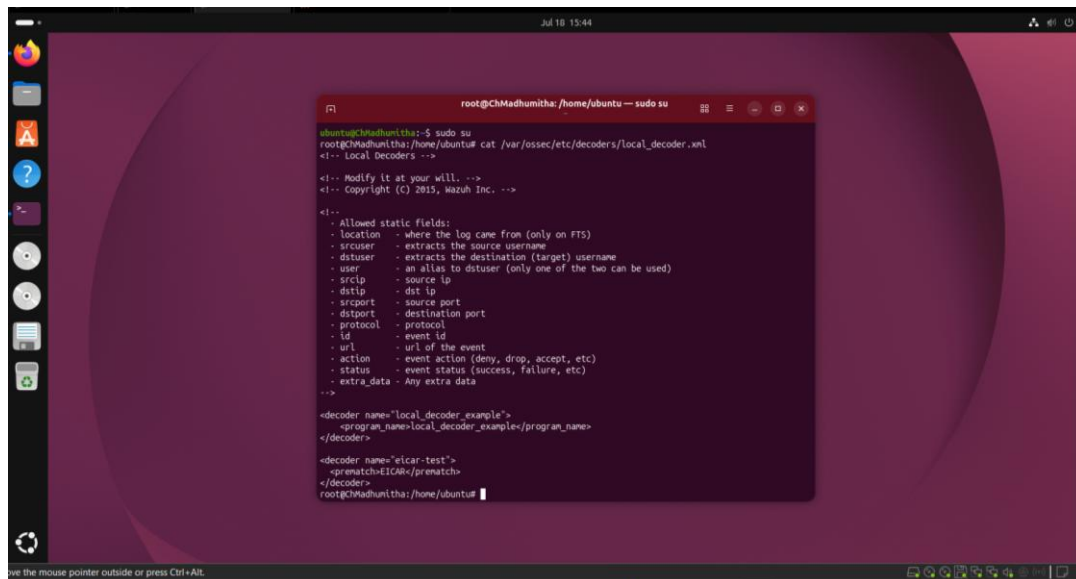


Figure 6: Custom Wazuh decoder configuration.

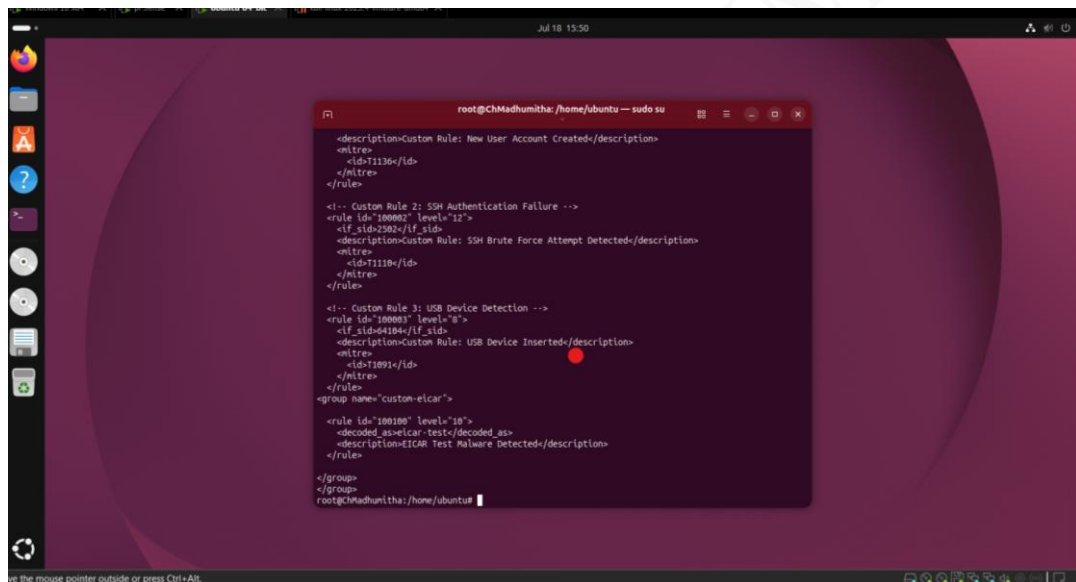


Figure 7: Custom Wazuh detection rule configuration.

10. Malware Detection Using the EICAR Test File

The EICAR test file was downloaded and executed in a controlled environment to safely simulate malware activity. ClamAV successfully detected the test file and prevented it from being treated as a normal file. The detection event was forwarded to Wazuh for centralized monitoring and analysis.

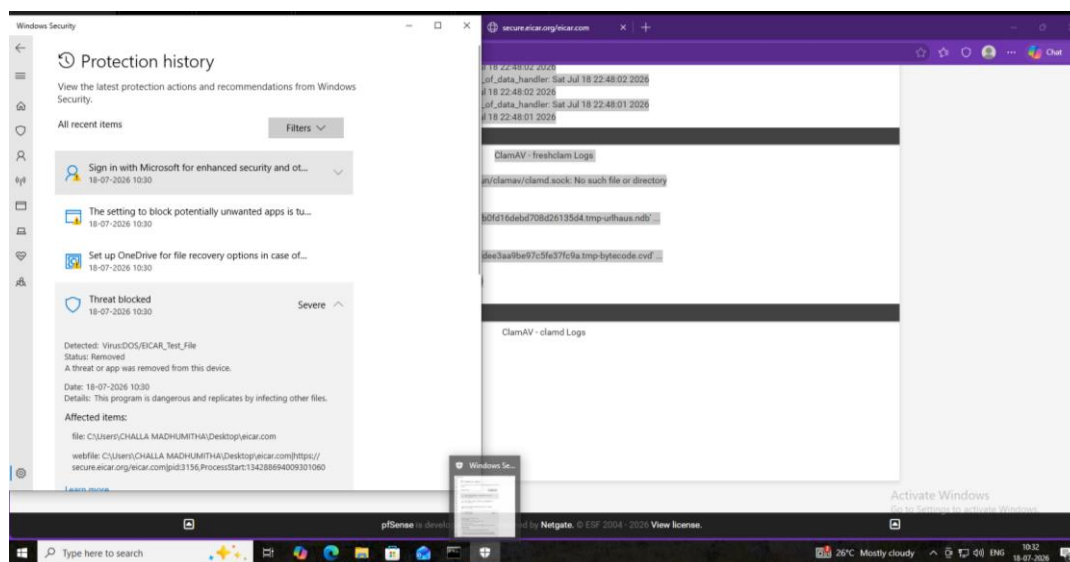


Figure 8: Successful detection of the EICAR test file.

11. Security Alert Monitoring

The Wazuh Dashboard was used to monitor alerts generated during malware testing. Security events were analyzed based on timestamps, rule IDs, alert severity, source system, and event details. This confirmed that Wazuh successfully detected and classified the malware-related activity.

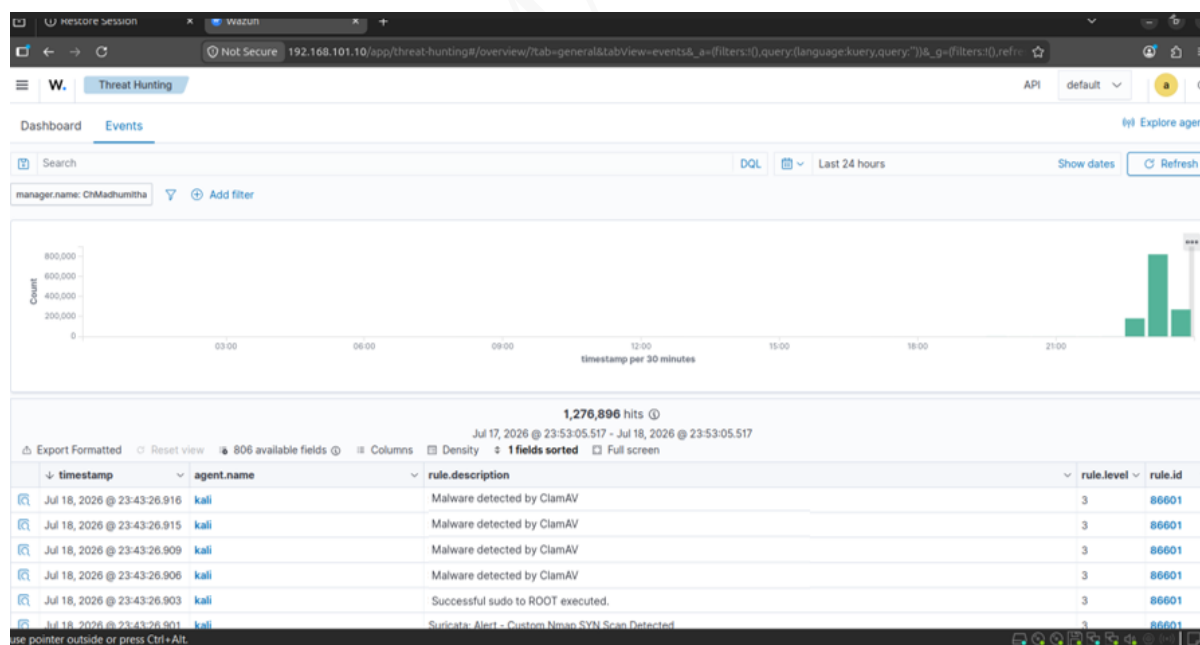


Figure 9: Wazuh Dashboard displaying malware detection alerts.

12. Incident Response Plan

Following malware detection, a structured Incident Response Plan was prepared to demonstrate the actions that should be taken during a real security incident.

Detection

Malware activity was identified through ClamAV scanning and Wazuh SIEM alerts.

Analysis

Security logs, alert severity, source IP addresses, affected hosts, and malware indicators were reviewed to determine the scope of the incident.

Containment

The affected endpoint was isolated from the network to prevent further spread of malicious activity.

Eradication

The detected malware file was removed, antivirus signatures were updated, and affected systems were scanned for additional threats.

Recovery

The endpoint was restored to normal operation after verifying that no further malicious activity remained.

Lessons Learned

Firewall rules, antivirus policies, and SIEM detection rules were reviewed and improved to strengthen future malware detection and incident response capabilities.

13. Results

The implementation was completed successfully.

Achievements:

- Successfully deployed pfSense Firewall.
- Installed and configured Squid Proxy.
- Integrated ClamAV with Squid Proxy.
- Configured Syslog-NG log forwarding.
- Integrated pfSense logs with Wazuh SIEM.
- Created custom decoders and detection rules.
- Successfully detected the EICAR malware test file.
- Generated and analyzed Wazuh security alerts.

- Prepared a complete Incident Response Plan.
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14. Learning Outcomes

Through this project, I gained practical knowledge in:

- Firewall administration using pfSense.
 - Proxy server deployment using Squid.
 - Malware detection using ClamAV.
 - Centralized log management using Syslog-NG.
 - SIEM monitoring with Wazuh.
 - Custom decoder and rule creation.
 - Security event analysis.
 - Incident response methodology.
 - Malware investigation within a SOC environment.
 - Blue Team security operations.
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15. Conclusion

This project provided hands-on experience in deploying an integrated security monitoring environment using pfSense Firewall, Squid Proxy, ClamAV, and Wazuh SIEM. By successfully detecting malware with the EICAR test file, monitoring security alerts, and preparing an Incident Response Plan, I developed practical skills in threat detection, log analysis, and incident handling. The project strengthened my understanding of Security Operations Center (SOC) workflows and demonstrated how multiple security technologies work together to detect, monitor, and respond to cyber threats.
